



Database Management Systems

Week 7, Day 2

MongoDB—Aggregate Functions

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Arrays in \$group

```
> db.persons.aggregate(  
  [  
    {$limit : 2},  
    {$group : {_id : "$tags"}}  
  ]  
< {  
  _id: [  
    'ut',  
    'voluptate',  
    'consequat',  
    'consequat'  
  ]  
}
```

- Splits each document with specified array to several documents - one document per array element

```
{ $unwind: <arrayReferenceExpression> }
```

\$unwind Stage

\$unwind

```
>_MONGOSH
```

```
> db.persons.aggregate([
  {$unwind: "$tags"},
  {$project: {_id:0, name:1, gender: 1, tags:1}}
])
< {
  name: 'Aurelia Gonzales',
  gender: 'female',
  tags: 'enim'
}
```

\$unwind and \$group

```
> db.persons.aggregate([
  {$unwind: "$tags"},
  {$group: {_id: "$tags"}},
])
< {
  _id: 'nisi'
}
{
  _id: 'in'
}
{
  _id: 'laborum'
}
```

\$sum

\$avg

\$max

\$min

Accumulator Operators



Accumulator syntax

- Accumulators maintain state for each group of the documents
`{ $<accumulatorOperator>: <expression> }`

\$sum with \$group

```
> db.persons.aggregate([
  {$group : {_id: "$favoriteFruit", count: {$sum: 1}}}
])
< {
  _id: 'banana',
  count: 339
}
{
  _id: 'apple',
  count: 338
}
{
  _id: 'strawberry',
  count: 323
}
```

Count people of Age

```
> db.persons.aggregate([
  {$group : {_id: "$age", count: {$sum: 1}}}
])
< {
  _id: 33,
  count: 50
}
{
  _id: 39,
  count: 65
}
```

\$sum

```
db.persons.aggregate([
  {$unwind: "$tags"},
  {$group : {_id: "$tags", count: {$sum: 1}}}]
)
{
  _id: 'in',
  count: 54
}
{
  _id: 'nisi',
  count: 48
}
```

Question(s)
